



Hepatic triacylglycerol associations with production and health in dairy cows

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ABSTRACT

The objectives were to evaluate the associations between hepatic triacylglycerol content and production, blood metabolites, incidence of diseases, reproduction, and survival in Holstein cows. Data were collected from 4 experiments including 329 cows in which hepatic tissue was sampled with a mean ($\pm$ standard deviation) of 8.3 ± 1.5 d postpartum (6 to 11 d) and analyzed for triacylglycerol concentration. The mean ($\pm$ standard deviation) concentration was $4.4 \pm 2.8\%$ on a wet basis and ranged from 0.4 to 16.1%. Intakes of dry matter (DM), energy balance, body weight (BW), body condition (BCS), productive performance, and incidence of diseases were evaluated for the first 105 d postpartum, whereas blood metabolites were assayed in the first 21 d postpartum. Reproductive performance and survival were monitored in the first 300 d postpartum. Mixed models were fitted to the data to investigate the linear and quadratic associations of hepatic triacylglycerol concentration with responses of interest. Increased concentration of hepatic triacylglycerol was associated with a quadratic increase in yields of milk, energy-corrected milk (ECM), and milk components. A change in hepatic triacylglycerol from 2.5 to 7.5% of the wet tissue was associated with an increase in yield of ECM of 1.8 kg/d, and with 0.2 kg more ECM per kg of DM intake. However, the increased efficiency was accompanied by decreases in DM intake, BCS, more exacerbated losses of BW, and a more negative body energy change. Increased concentration of hepatic triacylglycerol was associated with a quadratic increase in blood fatty acids and a linear increase in blood β -hydroxybutyrate concentrations, concurrent with linear decreases in concentrations of glucose and total Ca in blood. Moreover, a change in hepatic triacylglycerol from 2.5 to 7.5% was associated with linear increases in the relative risk of hyperketonemia by 2.5 times (15.2 vs. 37.5%), hypocalcemia by 1.7 times (30.3 vs. 52.4%), metritis by 2.1 times (12.5 vs. 25.7%), and diagnosis of multiple diseases postpartum by 2.4 times

(8.7 vs. 21.1%). Survival in the herd by 300 d postpartum tended to decrease from 91.1 to 86.3% with an increase in hepatic triacylglycerol from 2.5 to 7.5% of the wet tissue, but no association was observed between hepatic triacylglycerol and measures of reproduction in the first 300 d postpartum. Concentrations of hepatic triacylglycerol in early lactation varied substantially, and increments resulted in quadratic association with productive performance, but at the expense of tissue reserves as those cows had increased tissue catabolism and risk of diseases that reduced survival.

Key words: dairy cow, fatty liver, hepatic triacylglycerol, performance

INTRODUCTION

Dairy cows often experience hepatic lipidosis, characterized by increased hepatic triacylglycerol concentration above 3.0% of the tissue weight (Katoh, 2002), that usually peaks between 1 and 2 wk postpartum (Bollatti et al., 2020b). The increments in hepatic triacylglycerol are associated with negative nutrient balance that is typical of early lactation (Bobe et al., 2004), as the result of a rapid and extensive mobilization of body tissues, in particular fat depots, that results in increased hepatic uptake of circulating fatty acids (Drackley, 1999) and subsequent lipidosis of the hepatic tissue (Bobe et al., 2004). Fatty acids taken up by hepatocytes can be partially or completely oxidized (Kleppe et al., 1988), or in excess, they can be re-esterified to form triacylglycerols that are stored as lipid droplets into the hepatic parenchyma or exported as very low-density lipoproteins (Katoh, 2002). Ruminants are known to have reduced secretion of very low-density lipoproteins (Emery et al., 1992), which impairs hepatic export of triacylglycerols. Concurrent with negative nutrient balance, early lactation is characterized by increased risk of inflammatory diseases (Ribeiro et al., 2016), and disease induces conserved mechanisms involved in sickness behavior that includes anorexia (Brown and Bradford, 2021), further increasing lipolysis. Moreover, inflammatory mediators released during inflammation exacerbate hepatic accumulation of triacylglycerol independent of appetite (Bradford et al., 2009), thus increasing the risk of hepatic lipidosis.

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Reid (1980) reported that fat occupied 15 to 32% of the hepatic parenchyma in multiparous cows between wk 1 and 8 postpartum. Nevertheless, hepatic total fat content is not a good measure of lipidosis because of the high basal content of nontriacylglycerol lipids in bovine liver (Gaal et al., 1983), and that most of the increase in hepatic lipids during negative nutrient balance is caused by increments in the triacylglycerol fraction (Brumby et al., 1975). Gaal et al. (1983) assayed hepatic tissue sampled between 7 and 13 d postpartum for triacylglycerol content concurrent with stereological analysis to quantify the area of the hepatic parenchyma occupied by fat. The authors showed that triacylglycerol content was more correlated with the area occupied by lipids in hepatocytes than the total lipid content. Based on their findings and those of Reid (1980), they proposed an arbitrary classification of mild ($\leq 5\%$ triacylglycerol), moderate (5.1 to 10.0% triacylglycerol), or severe ($>10.0\%$ triacylglycerol) fatty liver (Gaal et al., 1983). Using Gaal et al. (1983) classification, Jorritsma et al. (2001) analyzed hepatic tissue sampled between 6 and 17 d postpartum from 218 dairy cows in 9 dairy herds and used the 5% threshold to classify cows as having fatty liver. The authors observed a positive association between hepatic triacylglycerol content and milk yield between 4 and 12 wk postpartum. The same group, then using 360 cows from the same 9 herds, showed that those with fatty liver, based on the 5% hepatic triacylglycerol threshold, had reduced rate of pregnancy (Jorritsma et al., 2000). Thus, the data from Jorritsma et al. (2000, 2001) suggested that more productive cows have increased triacylglycerol infiltration into the hepatic parenchyma and cows with hepatic triacylglycerol above 5% have compromised rate of pregnancy. Conversely, one could interpret their findings as hepatic triacylglycerol not having a detrimental effect on productive performance in early lactation and the reduced rate of pregnancy in cows with fatty liver being caused by a lesser rate of insemination as pregnancy per AI did not differ between cows with hepatic triacylglycerol below or above 5% (Jorritsma et al., 2000).

When cows are gaining body tissue, triacylglycerol represents less than 2% of the hepatic tissue weight (Brumby et al., 1975; Zenobi et al., 2018b), but it rapidly increases to above 5% on a wet basis during periods of intensive lipomobilization (Brumby et al., 1975; Zenobi et al., 2018b). Although hepatic lipidosis has been defined as triacylglycerol content greater than 3 (Kato, 2002) or 5% of the hepatic tissue (Gaal et al., 1983), those suggestions have been arbitrary and based primarily on the fact that well-fed cows have limited content of hepatic triacylglycerol. Thus, it is plausible that moderate increments in hepatic lipids might not

necessarily be detrimental to performance or health; however, it remains unclear what level of increment in hepatic triacylglycerol is associated with impaired lactation, health, or reproduction in dairy cows.

We hypothesized that concentrations of hepatic triacylglycerol in early lactation are associated with subsequent performance in Holstein cows and responses might not follow a linear association because small increases in hepatic triacylglycerol might not be detrimental to performance. Thus, the objective of this cohort study was to evaluate the relationships between early postpartum concentrations of hepatic triacylglycerol with production, concentrations of metabolites in blood, incidence of diseases, reproduction, and survival in dairy cows.

MATERIALS AND METHODS

Data from a convenience sample of experiments that collected hepatic tissue for triacylglycerol assay in the first 2 wk of lactation were collected for this retrospective cohort study (Lima et al., 2012; Zenobi et al., 2018a; Bollatti et al., 2020a,b; Zimpel et al., 2021a,b). In this study, experiment 1 refers to Zenobi et al. (2018a), 2 refers to Bollatti et al. (2020a,b), 3 refers to Zimpel et al. (2021a,b), and 4 refers to Lima et al. (2012). Details of cows, housing, diets, and general management are presented on the respective manuscripts. Inclusion criteria included dairy cows with hepatic tissue collected between 6 and 11 DIM that had available detailed records of production, health, and reproduction. A total of 329 Holstein cows contributed data to the study, 92 from Zenobi et al. (2018a), 95 from Bollatti et al. (2020a,b), 97 from Zimpel et al. (2021a,b), and 45 from Lima et al. (2012). Description of ingredient composition of diets, and chemical assays to analyze pre- and postpartum diets, were provided in the respective manuscripts (Lima et al., 2012; Zenobi et al., 2018a; Bollatti et al., 2020a; Zimpel et al., 2021a).

The present study used data from experiments with previously approved animal care protocols by the Institutional Animal Care and Use Committee (IACUC) of the respective universities. Therefore, the current study did not require additional approval by IACUC. Details of protocols approval are reported in the manuscripts of the original experiments.

Study Design and Sample Size

The study design was a retrospective cohort study. The exposure factor was the observed concentration of triacylglycerol in hepatic tissue sampled between 6 and 11 d postpartum. The sample size was a convenience sample based on the data from experiments in which

hepatic triacylglycerol was quantified in early lactation and detailed information on intake, production, health, and reproduction were collected from the study population. Investigators were blinded to the concentration of triacylglycerol detected in the hepatic tissue during data collection of cows because triacylglycerol was quantified in the hepatic tissue posterior to completion of each experiment and data collection.

Hepatic Tissue Collection

Hepatic tissue was collected once for biopsy on day (mean $\pm$ SD) 8.3 ± 1.5 (range from d 6 to 11) postpartum. Briefly, the cow was restrained, and the areas covering the 10th to 12th right intercostal spaces were scanned by ultrasonography (Aloka SSD-500V equipped with a 3.5-MHz convex transducer, Aloka Co. Ltd.) to determine the location for sampling hepatic tissue. The area was shaved and thoroughly cleansed with 3 alternating rounds of povidone-iodine scrubbing followed by a 70% isopropyl alcohol rinse. Upon disinfecting the surgical site, 15 mL of a solution containing 2% lidocaine hydrochloride was administered in the subcutaneous space and intercostal muscles. The surgical site was disinfected again as described previously. After a 2-cm incision of the skin, a stainless-steel percutaneous hepatic biopsy tool (Aries Surgical) was introduced aiming the hepatic tissue previously visualized by ultrasonography and approximately 1 g of tissue was collected. The tissue was placed in filter paper, rinsed with sterile saline, sliced into 3 sections, transferred into 3 separate cryovials, snap-frozen in liquid N₂, and stored at -80°C until analyses.

Analysis of Hepatic Tissue

Hepatic triacylglycerol was analyzed on liver samples from all 329 cows. One cryovial containing at least 300 mg of hepatic tissue was thawed on ice, and the sample transferred to filter paper to dry. A 300-mg sample was weighed on calibrated balance and homogenized in 3 mL of saline solution using an Ultra-Turrax T25 homogenizer (Rose Scientific Ltd.). Thereafter, a 6-mL solution of 1:1 saline and methanol was added to each sample and then split into 2 aliquots. One aliquot of 1.5 mL was used to analyze tissue DM and a second aliquot of 6.0 mL was used to analyze triacylglycerol. From this point on, all procedures were performed in triplicate. The DM of hepatic tissue was measured using the 1.5-mL aliquot of homogenate that was split into 3 equal aliquots of 0.5 mL. Each 0.5-mL aliquot was placed on a previously weighed aluminum pan. A fourth aluminum pan was used to determine the DM content of the 1 to 1 saline:methanol solution that was

used to extract the triacylglycerol from the hepatic tissue specimen. The aluminum pans were dried at 60°C for 48 h in a forced-air oven. After 48 h, the pans were weighed separately, and DM of the homogenate was calculated. First, we calculated the dry sample weight (step 1) using the weight of the dried sample with pan minus the pan weight. Then we calculated the dry saline:methanol solution weight (step 2) using the weight of the dried saline:methanol solution plus pan minus the pan weight. Finally, the sample DM weight was calculated as the value of step 1 minus the value of step 2.

The DM content of the sample was calculated as the sample DM weight/sample wet basis weight. The triacylglycerol content in hepatic tissue was assayed using the 6-mL aliquot that was split into 3 equal aliquots of 2 mL each. In each 2-mL homogenate, a 5-mL solution of 2 to 1 chloroform:methanol was used to extract the triacylglycerols (Folch et al., 1957). The tubes were dried under N₂ gas followed by a second extraction step with a 10-mL solution of 2 to 1 chloroform:methanol. The tubes were dried again under N₂ gas, followed by resuspension with 2 mL of propanol in each of the 3 tubes. Each of the 3 extracted samples were assayed for triacylglycerol concentration using a colorimetric method (Foster and Dunn, 1973). The concentration of triacylglycerol in the original sample of hepatic tissue was calculated using the concentration of triacylglycerol in the extracted sample and the amount of hepatic tissue used for the assay.

BCS and BW

Cows were scored for body condition once or twice weekly, from calving to 105 DIM, using a 1 to 5 scale with increments of 0.25 units as depicted in the Elanco BCS chart (Elanco Animal Health, 2009). In 3 of the 4 experiments, cows were weighed 2 to 4 times daily, after each milking, for the first 15 wk postpartum, on a walk-through (AfiWeigh, SAE Afikim) located on the exit lane of the milking parlor. The BCS and BW values for each cow were averaged into a single mean per cow postpartum.

Measurements of Yields of Milk and Milk Components

In experiments 1, 2, and 4, cows were milked twice daily, whereas in experiment 3 cows were milked 4 times daily. Milk yield was recorded automatically at each milking (AfiFlo milk meters, SAE Afikim) in experiments 1, 2, and 3, but twice monthly in experiment 4. Samples of milk were collected twice weekly, on Mondays and Wednesdays, during the morning

and afternoon milkings, in experiments 1, 2, and 3, and twice monthly in experiment 4, for the first 105 d postpartum for measurements of concentrations of fat, true protein, lactose, and somatic cells at the local DHI laboratory. The SCC was transformed to the SCS for statistical analysis according to the formula: $SCS = \text{Log}_{10}(\text{SCC}/12.5)/\text{Log}_{10}(2)$. Milk yield and composition from each milking was used to calculate the final concentrations and yields of milk components for each sampling day that were subsequently applied to the respective week postpartum. Yield of milk corrected for energy was calculated according to NRC (2001) as follows: $\text{ECM, kg/d} = [(0.3246 \times \text{milk yield}) + (12.86 \times \text{fat yield}) + (7.04 \times \text{protein yield})]$. Values for productive performance were averaged into a single mean per cow for the first 15 wk postpartum. The cumulative milk yield for the first 300 DIM also was collected from all cows. Cows that left the herd before 300 DIM because they died or were sold for slaughter contributed data on cumulative milk yield up to the day when they left the herd.

Measurements of DMI

In experiments 1, 2, and 3, each cow was randomly assigned to an individual feeding gate (American Calan Inc.) for measurements of DMI. Intake was recorded daily from calving to 105 DIM by measuring the amounts offered and refused daily, and adjusting for the DM content of the TMR measured at 105°C for 24 h. No individual cow DMI postpartum was available for cows in experiment 4.

Calculations of Net Energy Balance and Changes in Body Energy

Energy balance was calculated for cows in experiments 1, 2, and 3 using daily caloric intake from DMI and the energy content of the diets according to NRC (2001) using the NE_L system. The needs for maintenance were calculated based on the metabolic BW (NE_M , $\text{Mcal/d} = \text{BW}^{0.75} \times 0.08$). The NE_L required for milk synthesis was calculated according to yields of fat, protein, and lactose based on NRC (2001) as follows: $\{\text{milk yield} \times [(0.0929 \times \text{fat } \%) + (0.0563 \times \text{protein } \%) + (0.0395 \times \text{lactose } \%)]\}$.

Body energy that was retained or mobilized as body reserves was calculated from changes in BW (kg/d), loss or gain, and considering BCS (1 to 5) measurements to adjust the NE_L content in tissue gained or mobilized using an equation from NRC (2001) as follows: $\text{body energy change (Mcal/d)} = [(1.88 + 1.036 \times \text{BCS}) \times \text{change in BW}]$. Values for body energy change are expressed as Mcal/d of NE_L .

Blood Sampling

Blood sampled on d 7, 14, and 21 relative to parturition were assayed for concentrations of nonesterified fatty acids, BHB, and glucose. Blood collected on d 0, 1, and 3 postpartum was assayed for concentrations of total calcium (**tCa**). Details of laboratory assays for blood metabolites are presented in the respective manuscripts.

Health Responses and Survival

Diseases recorded and analyzed included milk fever, retained placenta, metritis, puerperal metritis, displaced abomasum, pneumonia, and mastitis for the first 105 d postpartum. Case definition and details of diagnosis are described in the original manuscripts. Morbidity included any of the clinical diseases investigated and cows with more than one clinical disease were classified as having multiple diseases.

Hyperketonemia was defined as plasma BHB concentration $>1.20 \text{ mM}$ on at least one blood sample collected on d 7, 14, and 21 postpartum (McArt et al., 2011). Incidence of subclinical hypocalcemia was based on at least one sample with plasma tCa $<2.0 \text{ mM}$ on d 0, 1, or 3 postpartum (Reinhardt et al., 2011). Survival of cows was evaluated for the first 300 d postpartum by recording cows that died or were sold.

Reproductive Responses

Details of the estrous or ovulation synchronization protocols have been presented in the original manuscripts. Briefly, cows had their ovulation synchronized for first AI in all 4 experiments. After the first AI, cows were observed daily for signs of estrus using tail chalk or pedometers and those in estrus were inseminated on the same day. Re-inseminated cows were considered as nonpregnant to the previous AI. Pregnancy was diagnosed by ultrasonography on d 32 after AI and cows diagnosed as nonpregnant were subjected to an ovulation synchronization protocol. Pregnant cows on d 32 were re-evaluated for pregnancy on d 60 of gestation, in experiment 4, or d 74 in experiments 1, 2, and 3. Throughout the manuscript, the reconfirmation of pregnancy will be labeled as d 74. Pregnancy was diagnosed based on the presence of an amniotic vesicle with an embryo with heartbeat. For statistical analyses of interval to pregnancy or rate of pregnancy, the diagnosis performed on d 74 after AI was used to determine whether a cow became pregnant in the study.

Cows became eligible to be inseminated once they passed the voluntary waiting period established in each experiment, which ranged from 65 ± 3 to 80 ± 3 d

postpartum. For calculations of 21-d cycle insemination rate and 21-d cycle pregnancy rate, it was assumed that a cow was eligible to be inseminated or become pregnant every 21 d passed the voluntary waiting period until she was diagnosed as pregnant on d 74, or until the farm coded her as do not inseminate, sold, or died within the first 300 DIM, whichever happened first.

The interval from calving to pregnancy was calculated using the same criteria previously described for 21-d cycle pregnancy rate. A cow was considered pregnant based on the diagnosis on d 74 after an AI, and those that remained as nonpregnant by 300 DIM or were coded as do not inseminate, sold, or died were censored using the day of the event that happened first within the first 300 DIM.

Statistical Analyses

All data were analyzed with linear mixed-effects models using SAS (version 9.4; SAS/STAT, SAS Institute Inc.). The mixed-effects models included the random effects of cohort, using treatment nested within experiment, and the interaction between cohort and hepatic triacylglycerol concentration. The rationale was to test the effect of hepatic triacylglycerol against the mean square of the interaction between cohort and hepatic triacylglycerol to be able to infer that significant associations were applicable to all treatments and experiments that originated the data used for this study, thus providing some evidence for external validity.

Continuous data were analyzed using the MIXED procedure of SAS (SAS/STAT, SAS Institute Inc.). The statistical models included the fixed effects of hepatic triacylglycerol concentration as linear and quadratic covariates, gestation length, calf sex category (male, female, or twins), days in the prepartum group, and the previous lactation 305-d milk yield deviation from the mean milk yield within experiment and parity group, and the random effects of treatment nested within experiment and the interaction between treatment nested within experiment and hepatic triacylglycerol concentration. For primiparous cows, the previous lactation milk yield was calculated as the genomic PTA for milk yield and using the base value for milk yield of the Holstein breed in 2015 obtained from the Council on Dairy Cattle Breeding (<https://www.uscdcb.com/>). Whenever the effect of the quadratic covariate of hepatic triacylglycerol resulted in $P > 0.10$, it was removed from the statistical model. All other covariates were retained in the final models. Distribution of residuals and homogeneity of variance were evaluated after fitting each statistical model. Concentration of BHB was transformed to the natural logarithm because residuals were not normally distributed. Nevertheless,

the significance of effects in the model did not change, nor did the interpretation of the result. To avoid using β coefficients and respective standard errors on a log scale, the results of the original data (untransformed) were used. The Kenward-Roger method was used to obtain the approximate denominator degrees of freedom for the F -tests of all statistical models.

Binary data such as incidence of diseases, survival, pregnancy per AI, pregnancy loss, 21-d insemination rate, and 21-d pregnancy rate were analyzed by logistic regression using the GLIMMIX procedure of SAS (SAS/STAT, SAS Institute Inc.) fitting a binary distribution. The models for responses with a single measurement per cow were exactly the same as those described for continuous variables. For responses with more than one measurement per cow such as pregnancy per AI for all AI, 21-d cycle insemination rate, and 21-d cycle pregnancy rate, the models also included the fixed effect of replicate (AI number or 21-d interval) and the random effect of cow. The quadratic covariate of hepatic triacylglycerol was removed from the statistical model whenever it resulted in $P > 0.10$. All other covariates were retained in the final models. The Kenward-Roger method was used to obtain the approximate denominator degrees of freedom for the F -tests of all statistical models. Dispersion of the data was assessed by the ratio of the generalized chi-squared statistic and its degrees of freedom (χ^2/df). In all models, except for pregnancy loss at second AI, the generalized χ^2/df was between 0.95 and 1.10, indicating that the variability in the data was properly modeled, and that only minor residual under- or overdispersion was present. Analyses of risk of milk fever and risk of pneumonia were not considered because of limited frequency of events that resulted in extreme underdispersion of the residual variance and poor predictability of the statistical models. Data are presented as least squares means and respective standard error of the mean.

The interval from calving to pregnancy according to different categories of hepatic triacylglycerol was analyzed with the Cox's proportional hazard regression using the PHREG procedure of SAS (SAS/STAT, SAS Institute Inc.). Cows were classified as having concentrations of hepatic triacylglycerol $<2.5\%$ ($n = 85$ cows), 2.5 to 4.99% ($n = 131$ cows), and $\geq 5.0\%$ ($n = 113$ cows) on d 8 postpartum. The statistical model included the fixed effects of category of hepatic triacylglycerol ($<2.5\%$, 2.5 to 4.99% or $\geq 5.0\%$), gestation length, calf sex category, days in the prepartum group, and the previous lactation milk yield deviation from the mean milk yield within experiment and parity group, and the random effects of treatment nested within experiment and the interaction between treatment nested within experiment and hepatic triacylglycerol category. The

adjusted hazard ratio and the 95% confidence interval were calculated. Survival curves were generated adjusted for covariates in the statistical model using the BASELINE statement in PHREG of SAS (SAS/STAT, SAS Institute Inc.). Model fit was assessed by plotting the Martingale and deviance residuals against the estimates of the linear predictor. Proportionality of hazards among treatments was evaluated using the ASSESS statement for proportional hazard in PHREG.

A linear mixed-effects model was used to obtain the predicted DMI for each cow by regressing DMI with major energy sinks such that residuals could be obtained to investigate if hepatic triacylglycerol is associated with residual DMI (**RFI**). The statistical model included fixed effects of NE_L secreted in milk, metabolic BW, body energy change, and parity group, and the random effect of treatment nested within experiment. Then, RFI (kg/d) was calculated as the difference between observed and predicted DMI and subsequently analyzed as a response variable as previously described.

Figures were created as scatter plots with the response of interest in the y-axis according to hepatic triacylglycerol content in the x-axis using the predicted responses for each cow contributing data for the statistical analysis of the dependent variable analyzed. The predicted values were obtained from the best linear unbiased estimates. Tables were created to discriminate the expected responses when hepatic triacylglycerol were 2.5, 5.0, or 7.5% of hepatic tissue on a wet basis using the LSM and SEM from the best linear unbiased estimates. The rationale for using the concentrations of 2.5, 5.0, and 7.5% hepatic triacylglycerol was to represent cows without hepatic lipidosis (2.5%), moderate degree of hepatic lipidosis (5.0%), or more severe hepatic lipidosis (7.5%) based on what has been described in the literature (Gaal et al., 1983; Katoh, 2002; Bobe et al., 2004), but still capture the responses within a range of hepatic triacylglycerol content that had a large number of observations such that the estimates represented what happened in the population of cows studied. Significance was declared when $P \leq 0.05$ and a tendency to differ at $0.05 < P \leq 0.10$.

RESULTS

A total of 329 cows contributed data for productive performance, blood metabolites, incidence of diseases, reproduction, and survival. Two hundred eighty-three cows contributed data for DMI, NE_L intake, and NE_L balance. The mean ($\pm$ SD) and median concentrations of hepatic triacylglycerol were, respectively, 4.37 ± 2.75 and 3.67% on a wet basis (ranged from 0.4 to 16.1%) and 12.02 ± 6.85 and 10.54% on a DM basis (ranged from 0.7 to 34.6%). A histogram of the concentration of

hepatic triacylglycerol is presented in Supplemental Figure S1 (<https://figshare.com/s/182fb6dbf0fc41ad223d>).

Production Performance

Concentration of hepatic triacylglycerol was associated quadratically ($P < 0.05$) with yields of milk, ECM, true protein, and lactose, whereas a tendency ($P < 0.10$) for a quadratic association was observed for yields of fat and cumulative milk by 300 d postpartum (Table 1). Yields of milk (Figure 1A), ECM (Figure 1B), and fat (Figure 1C), and cumulative milk by 300 DIM reached a plateau when hepatic triacylglycerol was approximately 6% of the wet weight, whereas yield of protein reached a plateau at 4% hepatic triacylglycerol (Figure 1D). Supplemental Table S1 (<https://figshare.com/s/182fb6dbf0fc41ad223d>) provides the β coefficients and respective standard error for the predictors in all final models for production performance of dairy cows. Increasing the concentration of hepatic triacylglycerol from 2.5 to 7.5% was associated with an increase in yields of milk by 2.1 kg/d, ECM by 1.8 kg/d, fat by 0.09 kg/d, true protein by 0.02 kg/d, and cumulative milk by 331 kg (Table 1). Increasing hepatic triacylglycerol resulted in a quadratic reduction in milk true protein content; however, no association was observed between hepatic triacylglycerol and milk fat content or SCS (Table 1).

Intake of Dry Matter, Measures of Energy Status and of Feed Efficiency

A negative linear association ($P < 0.05$) was observed between concentration of hepatic triacylglycerol and intakes of DM (Figure 2A) and NE_L , RFI, BW change, and body energy change (Table 1). Because of the reduced DMI and increased yield of ECM, the feed conversion ratio increased quadratically ($P < 0.001$) as hepatic triacylglycerol increased. In fact, hepatic triacylglycerol was linearly associated ($P < 0.001$) with RFI, reinforcing the improved efficiency in cows as hepatic triacylglycerol increased (Table 1). The peak feed conversion ratio was observed in cows with hepatic triacylglycerol of approximately 9% (Figure 2B). Nevertheless, the improvements in measures of efficiency were, to some extent, linked with increased losses of BW and BCS, reduced NE_L balance, and more body energy mobilization. Hepatic triacylglycerol was associated quadratically ($P \leq 0.02$) with NE_L balance and BCS, and it tended to be associated quadratically ($P = 0.06$) with the change in BCS postpartum (Table 2; Figures 2C and 2D). The β coefficients and respective SE for predictors retained in the final models for intake of DM, measures of energy status, and feed ef-

Table 1. Least squares means ($\pm$ SEM) for productive responses according to concentration of hepatic triacylglycerol on d 8 postpartum¹

Item ²	Hepatic triacylglycerol, % wet basis			P-value ³	
	2.5	5.0	7.5	Linear	Quadratic
Yield, kg/d					
Milk	41.1 $\pm$ 0.9	42.7 $\pm$ 1.0	43.2 $\pm$ 1.1	0.003	0.007
ECM	41.1 $\pm$ 0.9	42.5 $\pm$ 0.9	42.9 $\pm$ 1.1	0.01	0.03
Fat	1.51 $\pm$ 0.03	1.57 $\pm$ 0.04	1.60 $\pm$ 0.05	0.02	0.08
True protein	1.19 $\pm$ 0.03	1.21 $\pm$ 0.03	1.21 $\pm$ 0.04	0.03	0.02
Lactose	1.99 $\pm$ 0.05	2.08 $\pm$ 0.05	2.10 $\pm$ 0.05	0.001	0.008
Milk yield 300 d, ⁴ kg	10,312 $\pm$ 217	10,609 $\pm$ 267	10,643 $\pm$ 349	0.06	0.06
Content in milk, %					
Fat	3.74 $\pm$ 0.04	3.77 $\pm$ 0.04	3.80 $\pm$ 0.05	0.27	NR ⁵
True protein	2.93 $\pm$ 0.03	2.84 $\pm$ 0.03	2.79 $\pm$ 0.03	<0.001	0.002
Lactose	4.86 $\pm$ 0.02	4.84 $\pm$ 0.02	4.81 $\pm$ 0.02	0.04	NR
SCS	2.20 $\pm$ 0.21	2.35 $\pm$ 0.20	2.49 $\pm$ 0.23	0.12	NR
DMI, kg/d	22.0 $\pm$ 0.3	21.5 $\pm$ 0.3	21.1 $\pm$ 0.3	0.0004	NR
Residual DMI, kg/d	0.24 $\pm$ 0.23	-0.23 $\pm$ 0.26	-0.70 $\pm$ 0.33	<0.001	NR
Net energy intake, Mcal/d	36.7 $\pm$ 0.5	35.9 $\pm$ 0.5	35.2 $\pm$ 0.6	0.0004	NR
Net energy balance, Mcal/d	-2.07 $\pm$ 0.58	-4.05 $\pm$ 0.67	-5.12 $\pm$ 0.82	<0.0001	0.002
ECM/DMI, kg/kg	1.87 $\pm$ 0.03	2.01 $\pm$ 0.04	2.06 $\pm$ 0.05	<0.0001	<0.001
BW					
Mean, kg	644 $\pm$ 8	652 $\pm$ 12	660 $\pm$ 17	0.20	NR
Change, kg/d	-0.25 $\pm$ 0.05	-0.41 $\pm$ 0.05	-0.56 $\pm$ 0.07	<0.001	NR
BCS, 1 to 5					
Mean	3.20 $\pm$ 0.02	3.15 $\pm$ 0.03	3.14 $\pm$ 0.03	0.007	0.02
Change	-0.33 $\pm$ 0.05	-0.47 $\pm$ 0.05	-0.55 $\pm$ 0.06	<0.001	0.06
Body energy change, Mcal/d	-1.56 $\pm$ 0.28	-2.50 $\pm$ 0.28	-3.44 $\pm$ 0.39	<0.001	NR

¹Predicted responses of cows with triacylglycerol concentrations of 2.5, 5.0, or 7.5% of hepatic tissue on a wet basis.²Responses were evaluated in the first 105 d postpartum.³Linear = effect of the linear covariate of hepatic triacylglycerol concentration on a wet basis; quadratic = effect of the quadratic covariate of hepatic triacylglycerol concentration on a wet basis.⁴Cumulative milk yield by 300 d postpartum. Cows that left the herd before 300 DIM contributed data until the day they died or were sold.⁵NR = not retained in the final model because $P > 0.10$.

iciency are presented in Supplemental Table S1 (<https://figshare.com/s/182fb6dbf0fc41ad223d>).

Blood Metabolites and Subclinical Diseases

Hepatic triacylglycerol was associated quadratically ($P = 0.001$) with concentrations of nonesterified fatty acids, whereas the associations between hepatic triacylglycerol and concentrations of BHB, glucose, and tCa were all linear (Table 2; Figure 3A to 3D). As the concentration of hepatic triacylglycerol increased, blood fatty acids and BHB also increased ($P = 0.001$), whereas those of glucose and tCa decreased linearly ($P = 0.001$). An increment in concentrations of hepatic triacylglycerol from 2.5 to 7.5% was associated ($P < 0.01$) with increases in concentrations of fatty acids and BHB in blood of 0.24 and 0.39 mM, respectively (Table 2). The same change in hepatic triacylglycerol was associated with linear decreases ($P = 0.001$) in concentrations of glucose and tCa in blood of 0.22 and 0.11 mM, respectively. The risk of hyperketonemia and hypocalcemia increased linearly ($P < 0.001$) with increments in hepatic triacylglycerol. A change in hepatic triacylglycerol from 2.5 to 7.5% was associated with a

2.5- and 1.7-times increased risk of hyperketonemia and hypocalcemia, respectively (Table 2). The β coefficients and respective standard error for predictors retained in the final models for blood metabolites, and subclinical diseases are presented in Supplemental Table S2 (<https://figshare.com/s/182fb6dbf0fc41ad223d>).

Incidence of Clinical Diseases

Postpartum concentration of hepatic triacylglycerol was associated linearly ($P = 0.01$) with the risk of metritis and that of multiple diseases, and it tended to be associated ($P < 0.10$) with puerperal metritis and morbidity (Table 3). No association between hepatic triacylglycerol and risk of retained placenta, displaced abomasum, and mastitis was observed. The risk of milk fever and pneumonia were not statistically analyzed because of the low frequency of event that resulted in underdispersion of data and poor predictability of the statistical models. Three cases of milk fever were diagnosed in the study population, and they had 2.5, 6.9, and 12.5% hepatic triacylglycerol. Pneumonia affected 6 cows that had 3.1, 3.2, 3.4, 5.4, 6.9, and 16.1% hepatic triacylglycerol. The cases of milk fever and respiratory

disease contributed data for the analyses of morbidity and cows having multiple clinical diseases. The β coefficients and respective standard error for predictors retained in the final models for clinical diseases are presented in Supplemental Table S3 (<https://figshare.com/s/182fb6dbf0fc41ad223d>).

Reproduction and Survival

Of the 329 cows in the study, 319 received at least 1 AI and had a pregnancy diagnosis performed. The 10 cows that did not receive an insemination was because they were sold (5 cows), died (1 cow), or were

coded as do not inseminate (3 cows) before the end of the voluntary waiting period. The proportion of cows receiving at least 1 AI decreased linearly ($P = 0.05$) as hepatic triacylglycerol increased, and for cows with 2.5 or 7.5% hepatic triacylglycerol the difference was of 2.3 percentage units (Table 4). Cows were subjected to timed AI for first AI, and no association ($P = 0.95$) was observed between hepatic triacylglycerol and days at first AI, which had a mean of 75.8 d. Hepatic triacylglycerol tended to be associated ($P = 0.09$) with pregnancy per AI evaluated on d 32 after the first AI, but this association was no longer present when pregnancy was re-evaluated on d 74 after the first

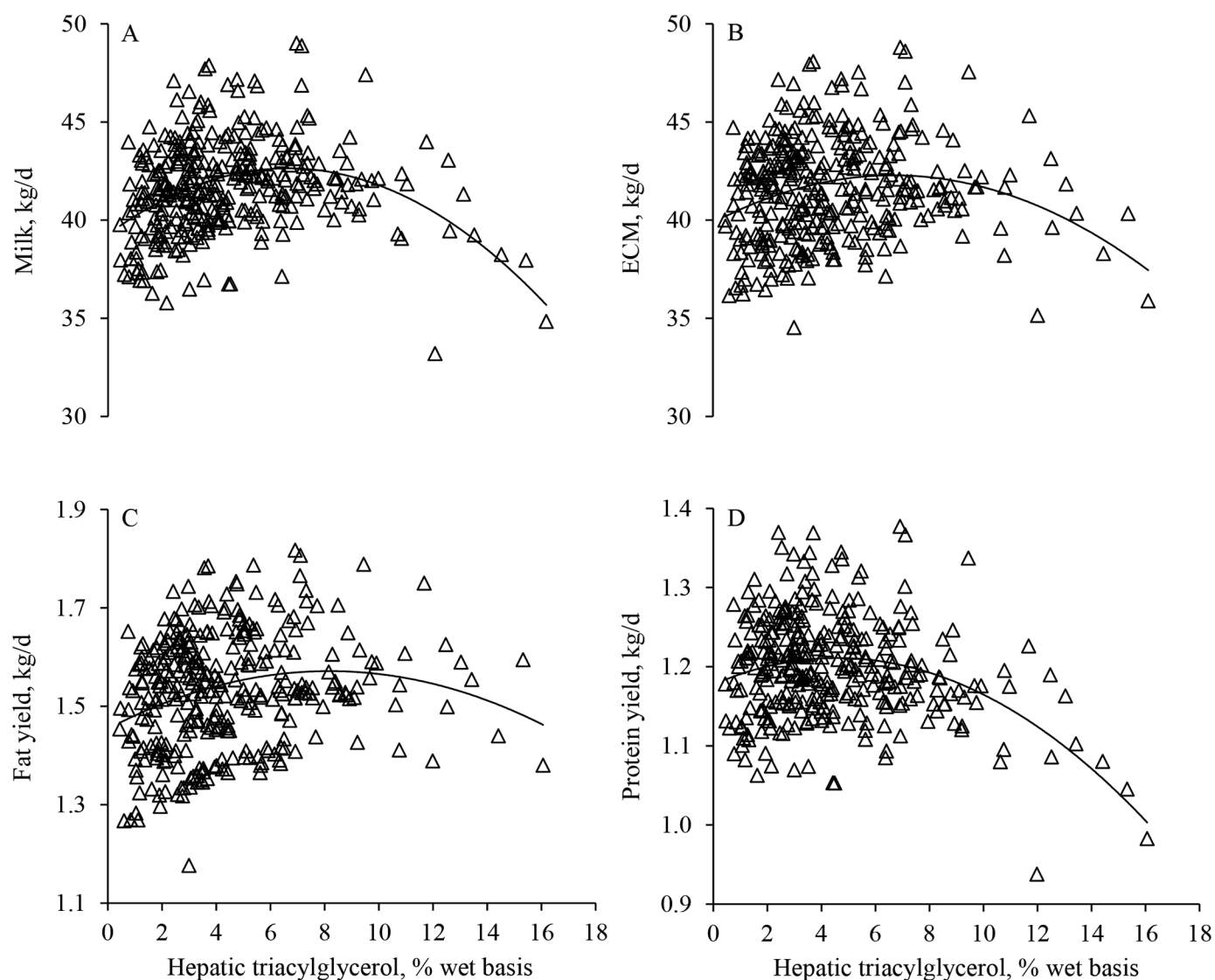


Figure 1. Yields of milk (A), ECM (B), fat (C), and true protein (D) in the first 105 d postpartum according to the concentrations of hepatic triacylglycerol. The continuous solid line, presented only when statistical significance was observed, depicts the response to concentrations of hepatic triacylglycerol. Panel A: linear ($P = 0.003$) and quadratic ($P = 0.007$) associations with hepatic triacylglycerol. Panel B: linear ($P = 0.01$) and quadratic ($P = 0.03$) associations with hepatic triacylglycerol. Panel C: linear ($P = 0.02$) and quadratic ($P = 0.08$) associations with hepatic triacylglycerol. Panel D: linear ($P = 0.03$) and quadratic ($P = 0.02$) associations with hepatic triacylglycerol.

AI (Table 4; Supplemental Table S4, <https://figshare.com/s/182fb6dbf0fc41ad223d>). In fact, no associations between the linear or quadratic covariates of hepatic triacylglycerol were observed with pregnancy per AI at all AI, insemination rate, pregnancy rate, or the proportion of pregnant cows by 300 DIM. When cows were categorized as having concentrations of hepatic triacylglycerol <2.5%, 2.5 to 4.99%, or $\geq 5.0\%$, the daily hazard of pregnancy was undistinguishable between the 3 groups (Figure 4).

Thirty-six of the 329 cows enrolled in the study left the herd by 300 DIM because of death or culling, a removal of 10.9%. Hepatic triacylglycerol tended to be associ-

ated linearly ($P = 0.10$) with survival by 300 DIM, and as hepatic triacylglycerol increased the risk of removal from the herd also increased (Supplemental Table S4, <https://figshare.com/s/182fb6dbf0fc41ad223d>). A change in concentration of hepatic triacylglycerol from 2.5 to 7.5% resulted in 1.54 times greater risk of herd removal by 300 DIM (Table 4).

DISCUSSION

During the first weeks of lactation, dairy cows experience extensive mobilization of body fat resulting in increased uptake of fatty acids by the liver (Reynolds

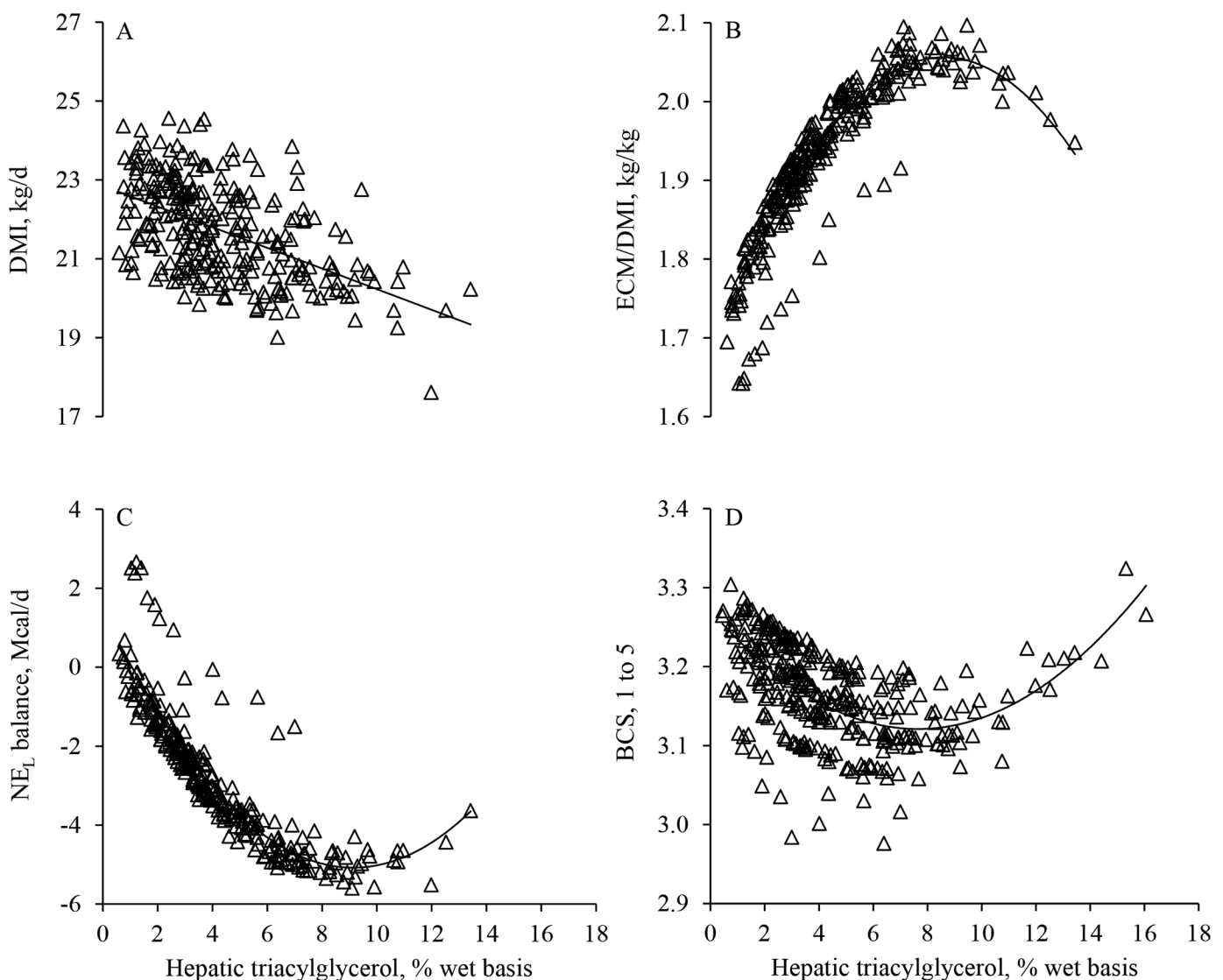


Figure 2. Dry matter intake (A), yield of ECM per kg of DMI (B), net energy balance (C), and BCS (D) in the first 105 d postpartum according to the concentrations of hepatic triacylglycerol. The continuous solid line, presented only when significance was observed, depicts the response to concentrations of hepatic triacylglycerol. Panel A: linear association with hepatic triacylglycerol ($P = 0.0004$). Panel B: linear ($P < 0.0001$) and quadratic ($P < 0.001$) associations with hepatic triacylglycerol. Panel C: linear ($P < 0.0001$) and quadratic ($P = 0.002$) associations with hepatic triacylglycerol. Panel D: linear ($P = 0.007$) and quadratic ($P = 0.02$) associations with hepatic triacylglycerol.

et al., 2003), and concurrent with limited export of triacylglycerol-rich lipoproteins (Emery et al., 1992), the hepatic tissue becomes enriched with triacylglycerols leading to the development of fatty liver (Bobe et al., 2004), which is thought to have detrimental effects on performance and health of dairy cows. In the present study, hepatic triacylglycerol was associated quadratically with productive performance. As the concentration of triacylglycerol in hepatic tissue increased, yields of milk and milk components increased and plateaued at concentrations ranging between 4 and 7%, after which production began to decline. In contrast, the association between hepatic triacylglycerol and metritis was linear, and risk of morbidity and of multiple diseases increased as hepatic triacylglycerol increased. Hepatic triacylglycerol was associated linearly with reduced DMI, but quadratically with increased feed conversion ratio, and the latter improved at the expense of body reserves.

Hepatic Triacylglycerol and Production Performance

The relationship between hepatic lipidoses and postpartum performance has not been extensively characterized in dairy cows. Cows in the well-fed state have limited hepatic triacylglycerol content (Brumby et al., 1975; Zenobi et al., 2018b), which rapidly increases to values above 5% on a wet basis during feed restriction or intensive lipomobilization (Brumby et al., 1975; Zenobi et al., 2018b). Because hepatic triacylglycerol increases with reduced DMI or energy balance, and the latter are typically linked with depressed performance; thus, it is not surprising that increments in hepatic lipid content are often associated with detrimental effects on health

and performance in dairy cows (Bobe et al., 2004). One of the challenges of this cohort study is to distinguish an effect of increased accumulation of hepatic triacylglycerol that might affect performance from that of negative nutrient balance that might have predisposed cows to develop increased accumulation of hepatic triacylglycerol and can potentially affect performance. It is noteworthy that as the hepatic triacylglycerol increased, DMI in the first 15 wk of lactation decreased; however, yield of ECM showed a quadratic association with hepatic triacylglycerol. It initially increased, reached a plateau at 4 to 7% hepatic triacylglycerol, and then declined. The increased yield of ECM in light of less DMI resulted in increased feed conversion ratio, which was partially explained by the increased loss of BW and BCS. When we calculated the NE_L balance in cows with 2.5 or 7.5% hepatic triacylglycerol, the difference between the 2 groups of cows amounted to 3.05 Mcal/d less for cows with 7.5% hepatic triacylglycerol. At the same time, when we calculated the differential change in daily body energy change, it supplied an estimated 1.88 Mcal more for cows with 7.5% hepatic triacylglycerol relative to cows with 2.5% triacylglycerol. The energy supplied by additional body tissue mobilization represented 61.6% ($1.88/3.05$) of the difference in NE_L balance for the cows with 7.5% hepatic triacylglycerol content. This implies that cows with increased hepatic triacylglycerol either had less maintenance requirements than calculated using current NRC (2001) guidelines or that the ability to use nutrients and extract calories from the diets differed from that of cows with less hepatic triacylglycerol.

The concentration of hepatic triacylglycerol that might affect cow health and performance has not been

Table 2. Least squares means ($\pm$ SEM) for blood metabolites and incidence of subclinical diseases according to concentration of hepatic triacylglycerol on d 8 postpartum¹

Item	Hepatic triacylglycerol, % wet basis			P-value ²	
	2.5	5.0	7.5	Linear	Quadratic
Blood metabolite, ³ mM					
Fatty acids	0.47 $\pm$ 0.02	0.61 $\pm$ 0.03	0.71 $\pm$ 0.04	<0.0001	0.001
BHB	0.73 $\pm$ 0.05	0.88 $\pm$ 0.07	1.04 $\pm$ 0.09	0.001	NR ⁴
Glucose	3.46 $\pm$ 0.10	3.35 $\pm$ 0.09	3.24 $\pm$ 0.10	<0.0001	NR
Total Ca	2.28 $\pm$ 0.05	2.23 $\pm$ 0.05	2.17 $\pm$ 0.05	0.001	NR
Subclinical disease, %					
Hyperketonemia ⁵	15.2 $\pm$ 3.1	24.7 $\pm$ 3.8	37.5 $\pm$ 5.6	<0.0001	NR
Hypocalcemia ⁶	30.3 $\pm$ 6.5	40.8 $\pm$ 7.1	52.4 $\pm$ 8.1	0.0003	NR

¹Predicted responses of cows with triacylglycerol concentrations of 2.5, 5.0, or 7.5% of hepatic tissue on a wet basis.

²Linear = effect of the linear covariate of hepatic triacylglycerol concentration on a wet basis; quadratic = effect of the quadratic covariate of hepatic triacylglycerol concentration on a wet basis.

³Samples collected and analyzed on d 7, 14, and 21 postpartum for fatty acids, BHB, and glucose, and on d 0, 1, and 3 postpartum for total Ca.

⁴NR = not retained in the final model because $P > 0.10$.

⁵Based on at least one plasma sample with BHB >1.20 mM on d 7, 14, or 21 postpartum.

⁶Based on at least one plasma sample with total Ca <2.0 mM on d 0, 1, or 3 postpartum.

critically scrutinized. Veenhuizen et al. (1991) assigned dairy cows to be fed 1,3-butanediol and subjected to feed restriction to 70% of that of ad libitum fed cows from 14 to 44 d postpartum. The authors observed that feed restriction enhanced lipolysis, which was evident based on the large increment in concentrations of plasma fatty acids and BHB and concurrent increased loss of BW. Cows accumulated hepatic triacylglycerols, from 3.5 to 9.5% on a wet basis between 9 and 44 d postpartum. The treatment imposed resulted in a gradual decline in milk yield during the period of dietary restriction compared with cows fed ad libitum, likely related to the reduced supply of nutrients but, perhaps, also as-

sociated with the potential damage the increase hepatic infiltration of lipid droplets might incur in dairy cows (Bobe et al., 2004). In the current study, increments in hepatic triacylglycerol were associated quadratically with productive performance, suggesting that a moderate hepatic lipidosis was not detrimental to lactational performance. In contrast, excessive triacylglycerol content in the hepatic tissue, linked to exacerbated tissue catabolism, was associated with depressed production, health, and survival in dairy cows.

It is well described that triacylglycerol accumulates in the liver of dairy cows in periods of negative nutrient balance following extensive lipomobilization (Bobe et

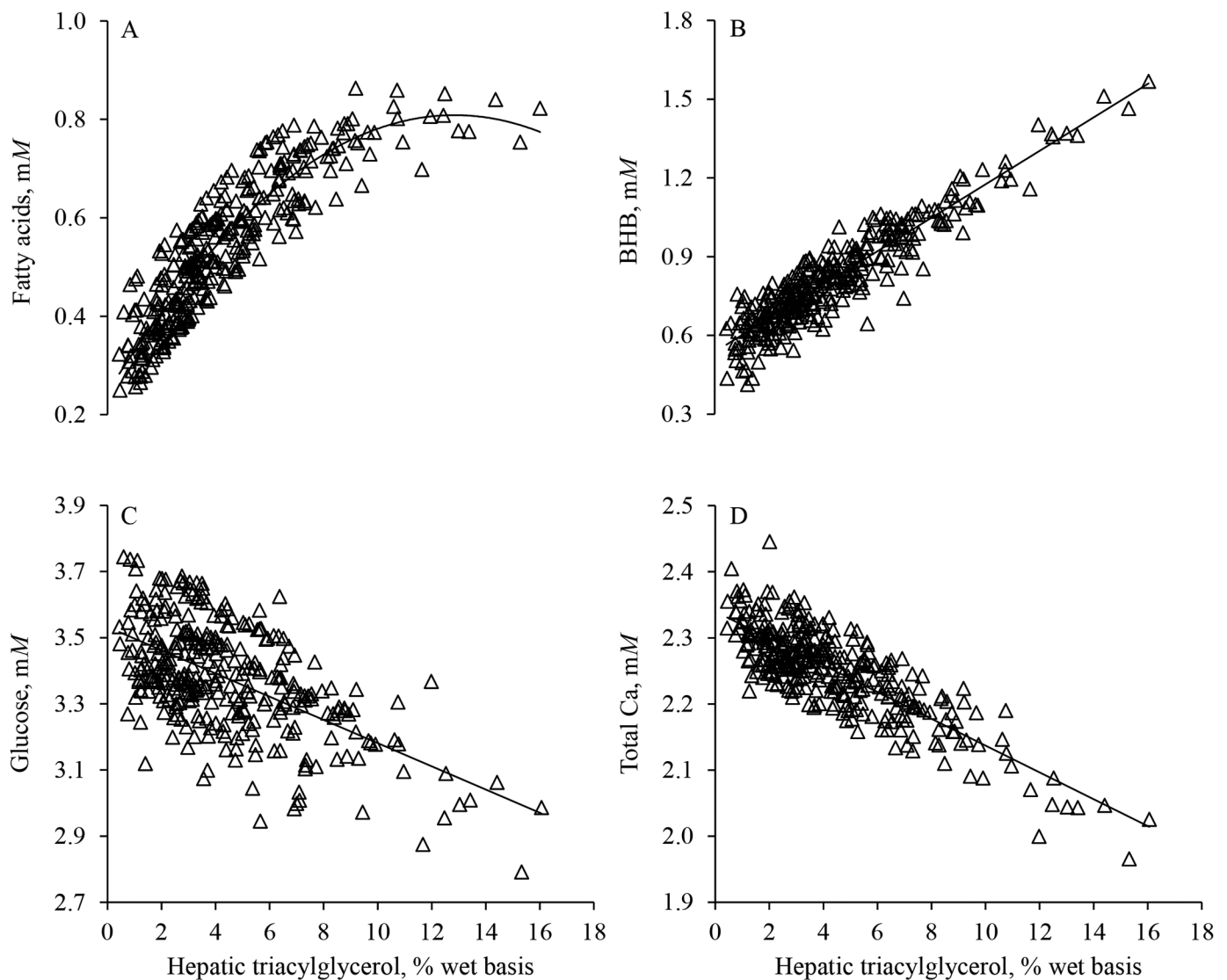


Figure 3. Concentrations of blood fatty acids (A), BHB (B), and glucose (C) in the first 21 d postpartum, and those of total Ca (D) in the first 3 d postpartum according to the concentrations of hepatic triacylglycerol. The continuous solid line, presented only when significance was observed, depicts the response to concentrations of hepatic triacylglycerol. Panel A: linear ($P < 0.0001$), and quadratic ($P = 0.001$) associations with hepatic triacylglycerol. Panel B: linear ($P = 0.001$) association with hepatic triacylglycerol. Panel C: linear ($P < 0.0001$) association with hepatic triacylglycerol. Panel D: linear ($P = 0.001$) association with hepatic triacylglycerol.

Table 3. Least squares means ($\pm$ SEM) for incidence of clinical diseases according to concentration of hepatic triacylglycerol on d 8 postpartum¹

Clinical disease, %	Hepatic triacylglycerol, % wet basis			P-value ²	
	2.5	5.0	7.5	Linear	Quadratic
Retained placenta	9.3 $\pm$ 2.7	11.9 $\pm$ 3.2	15.1 $\pm$ 4.7	0.12	NR ³
Metritis	12.5 $\pm$ 3.4	18.2 $\pm$ 4.5	25.7 $\pm$ 6.8	0.01	NR
Puerperal metritis	6.9 $\pm$ 2.3	10.2 $\pm$ 3.4	14.7 $\pm$ 6.0	0.07	NR
Displaced abomasum	1.0 $\pm$ 0.9	1.0 $\pm$ 0.8	1.0 $\pm$ 0.9	0.99	NR
Mastitis	14.2 $\pm$ 3.2	16.9 $\pm$ 3.3	19.9 $\pm$ 4.5	0.15	NR
Morbidity ⁴	36.2 $\pm$ 5.0	41.3 $\pm$ 5.4	46.7 $\pm$ 6.9	0.10	NR
Multiple diseases ⁵	8.7 $\pm$ 2.9	13.7 $\pm$ 4.1	21.1 $\pm$ 6.6	0.01	NR

¹Predicted responses of cows with triacylglycerol concentrations of 2.5, 5.0, or 7.5% of hepatic tissue on a wet basis.

²Linear = effect of the linear covariate of hepatic triacylglycerol concentration on a wet basis; quadratic = effect of the quadratic covariate of hepatic triacylglycerol concentration on a wet basis.

³NR = not retained in the final model because $P > 0.10$.

⁴Morbidity included retained placenta, metritis, displaced abomasum, milk fever, pneumonia, and mastitis.

⁵Multiple diseases included cows diagnosed with more than one type of disease listed in morbidity.

al., 2004). It is usually assumed that increases in hepatic triacylglycerol content is detrimental to performance because lipidosis has been linked with dysfunction of hepatocytes resulting in reduced ability to detoxify ammonia to urea (Strang et al., 1998), and reduced capacity to convert propionate to glucose (Veenhuizen et al., 1991; Overton et al., 1999). It is difficult to isolate the effects of hepatic triacylglycerol from those of negative nutrient balance, which often results in fatty liver, but is also linked with milk yield. In an attempt to minimize differences in potential for milk yield in cows with diverging hepatic triacylglycerol content, the centered value of 305-d milk yield observed in the lactation previous to the study was included as a covariate in the statistical models. The rationale was that milk yield would be adjusted to the mean potential for production based on the previous lactation. In spite of that, hepatic triacylglycerol was quadratically associ-

ated with yield of ECM, thus suggesting that small increments were not necessarily detrimental to productive performance. In corroboration with our findings, Jorritsma et al. (2001) collected liver tissue from 218 cows between 6 and 17 d postpartum across 9 Dutch dairy herds. The authors observed that increments in hepatic triacylglycerols were associated with a linear increase in milk yield. Nevertheless, from the present study, it is clear though that excessive concentrations of hepatic triacylglycerol, larger than 7% of the hepatic parenchyma, are linked with reduced yields of milk and milk component.

Hepatic Triacylglycerol and Mobilization of Body Tissue

Dairy cows have been selected in favor of milk production, but not necessarily for parallel increases in DMI in

Table 4. Least squares means ($\pm$ SEM) for reproduction and survival according to concentration of hepatic triacylglycerol on d 8 postpartum¹

Item ²	Hepatic triacylglycerol, % wet basis			P-value ³	
	2.5	5.0	7.5	Linear	Quadratic
Inseminated, %	98.8 $\pm$ 0.8	98.0 $\pm$ 1.2	96.5 $\pm$ 2.1	0.05	NR ⁴
Pregnancy at first AI, %					
d 32	39.4 $\pm$ 4.5	35.8 $\pm$ 4.1	32.5 $\pm$ 5.1	0.09	NR
d 74	37.1 $\pm$ 6.1	34.9 $\pm$ 5.7	32.9 $\pm$ 6.6	0.46	NR
Pregnancy loss	6.6 $\pm$ 3.9	3.7 $\pm$ 2.4	2.1 $\pm$ 2.1	0.25	NR
Pregnancy per AI at all AI, ⁵ %	29.4 $\pm$ 3.8	28.3 $\pm$ 3.6	27.2 $\pm$ 3.9	0.46	NR
21-d insemination rate, %	70.1 $\pm$ 3.2	69.3 $\pm$ 3.1	68.5 $\pm$ 3.4	0.51	NR
21-d pregnancy rate, ⁵ %	20.2 $\pm$ 3.6	19.1 $\pm$ 3.4	18.0 $\pm$ 3.5	0.27	NR
Pregnant by 300 DIM, %	77.1 $\pm$ 4.2	75.4 $\pm$ 4.9	73.6 $\pm$ 7.2	0.58	NR
Left the herd by 300 DIM, %	8.9 $\pm$ 1.9	11.0 $\pm$ 1.8	13.7 $\pm$ 2.8	0.10	NR

¹Predicted responses of cows with triacylglycerol concentrations of 2.5, 5.0, or 7.5% of hepatic tissue on a wet basis.

²Reproduction evaluated in the first 300 DIM or until the cow was diagnosed pregnant on d 74 after any AI or left the herd, whichever happened first.

³Linear = effect of the linear covariate of hepatic triacylglycerol concentration on a wet basis; quadratic = effect of the quadratic covariate of hepatic triacylglycerol concentration on a wet basis.

⁴NR = not retained in the final model because $P > 0.10$.

⁵Based on diagnosis performed on d 74 after each insemination.

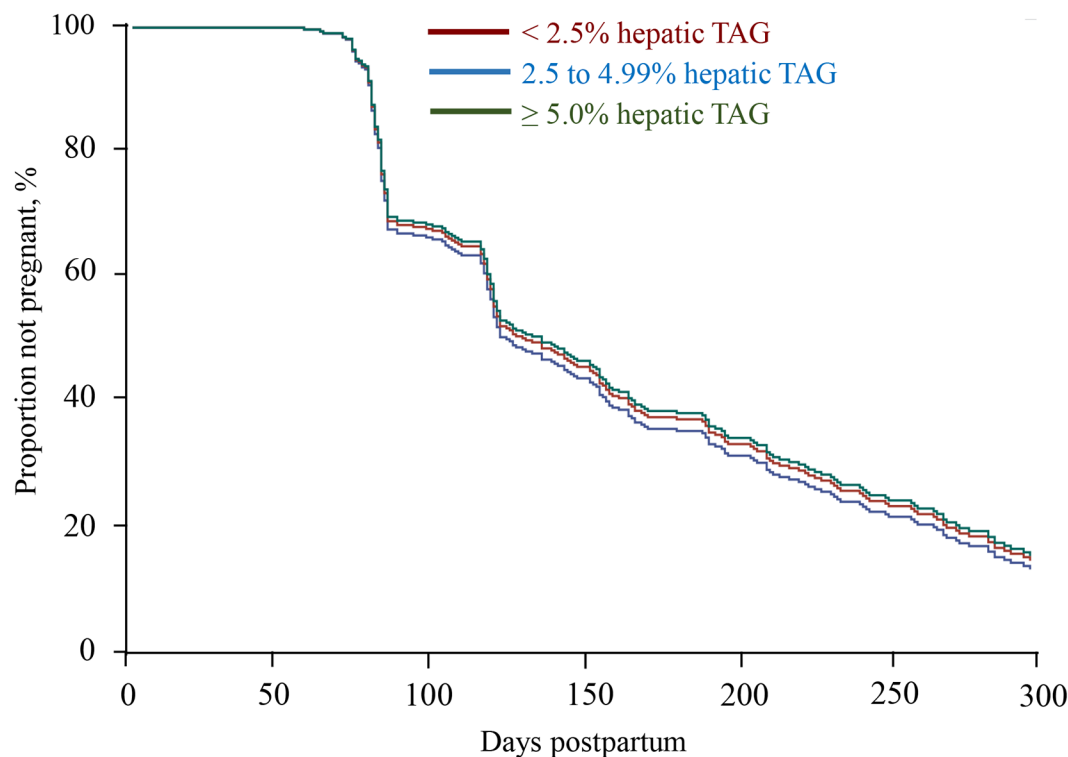


Figure 4. Survival curves for interval from calving to pregnancy in Holstein cows classified according to category of hepatic triacylglycerol (TAG) as TAG <2.5% (n = 85 cows), TAG = 2.5 to 4.99% (n = 131 cows), or TAG ≥5.0% (n = 113 cows) on d 8 postpartum. Category of TAG was not associated ($P = 0.87$) with the rate of pregnancy. The adjusted hazard ratios (95% CI) were 1.04 (0.76 to 1.43) for cows with TAG <2.5%, and 0.97 (0.70 to 1.34) for cows with TAG = 2.5 to 4.99% compared with cows with TAG ≥5.0%.

the first weeks of lactation (Miglior et al., 2017). Such dynamics of production and intake in early lactation are anticipated to result in adipose tissue catabolism. Indeed, cows with increased hepatic triacylglycerol had associated greater losses of BW and BCS that resulted in increased mobilization of body energy to support lactation. The observed improved productive performance and feed conversion ratio in cows with increased hepatic triacylglycerol was in detriment of body tissue reserves with signatures of increased lipomobilization. As hepatic triacylglycerol concentration increased, so did the concentrations of fatty acids and BHB. Mobilized fatty acids supply lipid substrate for the mammary gland for synthesis of milk fat, but they can also be extracted by extramammary tissues, including the liver, to undergo β -oxidation and ketogenesis (Reynolds et al., 2003). Although moderate lipomobilization is associated with benefits to postpartum productive performance, it can also be linked with more exacerbated negative NE_L balance and potentially increase the risk of metabolic and infectious diseases (Goff and Horst, 1997). Indeed, increased concentrations of hepatic triacylglycerol was associated linearly with increased risks of hyperketonemia, hypocalcemia, metritis, morbidity, and multiple diseases. The present study does not allow

us to establish any causal relationship, but it is possible that accumulation of lipid droplets in the hepatic parenchyma alters hepatocyte function and further compromises health of cows. It is important to note that the additional calories provided by the changes in body energy as hepatic triacylglycerol increased were not enough to explain the changes in NE_L balance, thus suggesting that either the energy released by the extra BW loss or the efficiency of use of dietary nutrients differed in cows with diverging concentrations of hepatic triacylglycerol.

Hepatic Triacylglycerol and Blood Metabolites

It is clear that cows with hepatic lipidosis have increased concentrations of fatty acids in blood and associated ketogenesis as observed herein; however, it is interesting to note that increased concentrations of hepatic triacylglycerol were associated linearly with decreased concentrations of tCa in blood in the first days postpartum. Using suggested threshold values for BHB and tCa, it was shown that as hepatic triacylglycerol increased, so did the risk of hyperketonemia and hypocalcemia. Concurrent, as hepatic triacylglycerol increased, so did the risk of morbidity, in particular

metritis. Diseases that occur in early lactation suppress appetite (Pérez-Báez et al., 2019a,b), with conserved mechanisms across multiple species (Brown and Bradford, 2021). Obviously, the reduced DMI and associated inflammatory response could have exacerbated hepatic triacylglycerol accumulation. On the reverse, cows predisposed to increased hepatic triacylglycerol might be more susceptible to diseases because lipid deposition into the hepatic parenchyma can be detrimental to cellular function and the liver plays a central role on metabolism and health of animals. Both reduced concentrations of tCa and increased concentrations of BHB and fatty acids in early lactation, which were observed as hepatic triacylglycerol increased, have been linked with risk of diseases in dairy cows (Chapinal et al., 2011). The link presumably is partly related to the role Ca plays in innate immune function (Martinez et al., 2014), and the disruptive effects BHB and fatty acids might have on immune cells (Hoeben et al., 1997; Lacetera et al., 2004). Therefore, it is plausible to suggest that cows having increased hepatic lipidosis suffer from underlying subclinical problems that altogether predispose them to morbidity.

Hepatic Triacylglycerol and Health

Cows that develop fatty liver usually undergo more extensive lipomobilization linked with inadequate DMI in late gestation and early lactation. A link exists between DMI or energy balance during the transition period and risk of peripartum diseases (Pérez-Báez et al., 2019a,b). As cows mobilized fat depots, the lipid composition of plasma fatty acids and phospholipids become enriched with palmitic and stearic acid (Contreras et al., 2010). Both of these SFA are known for their proinflammatory characteristics and can promote the synthesis of inflammatory cytokines such as TNF α and IL-1 β to trigger positive acute phase response in dairy cows (Horst et al., 2021). Cows induced to develop fatty liver have increased concentrations of acute phase proteins and other proinflammatory markers in early lactation (Ametaj et al., 2005), and such molecules are known to induce triacylglycerol accumulation in the bovine liver (Bradford et al., 2009). It is possible that such changes in lipid profiles might predispose cows to increased risk of morbidity (Chapinal et al., 2011).

As hepatic triacylglycerol increased, so did the risk of morbidity and multiple diseases. Obviously, this study does not allow us to identify the underlying causal relationship, if what might increase the risk of morbidity also increases hepatic triacylglycerol, or if indeed increased hepatic triacylglycerol content predisposes cows

to diseases. Accumulation of lipids into the hepatic parenchyma has been linked with liver dysfunction that might predispose animals to diseases. During hepatic lipidosis, triacylglycerols accumulate in organelles of cells, initially thought to be storage sites for lipids (Mashek et al., 2015); however, recent data show that hepatic lipid droplets are dynamic organelles and have active metabolism. These lipid droplets grow in size because of incorporation of additional lipids, coalesce to form very large droplets, reduce in size concurrent with lipolysis, and export as a component of lipoproteins (Gluchowski et al., 2017). However, accumulation of lipids causes cellular stress and organelle dysfunction that might impair important hepatic functions for ruminants such as gluconeogenesis and ureagenesis (Strang et al., 1998), which could affect health. Furthermore, hepatic lipidosis seems to induce a proinflammatory response at the same time that mediators of that response might further increase hepatic triacylglycerol accumulation (Bradford et al., 2009) and, perhaps, justify increased disease risk.

Hepatic Triacylglycerol and Reproductive Performance

It is clear that increased hepatic triacylglycerol is associated with tissue catabolism in dairy cows. For instance, as hepatic triacylglycerol increased from 2.5 to 7.5%, the daily mobilized body energy more than doubled, from 1.56 to 3.44 Mcal/d in the first 105 DIM. In humans, hepatic lipidosis is associated with insulin resistance (Gluchowski et al., 2017), which would favor adipose tissue lipolysis and increased concentrations of fatty acids and BHB in blood, as observed herein. Although a certain degree of insulin resistance likely benefits partition of nutrients to milk synthesis, it might have consequences to health, thus potentially justifying increments in morbidity as hepatic triacylglycerol increased. Also, as the degree of hepatic lipidosis rose, survival of cows tended to decrease. To our surprise, no association was observed between hepatic triacylglycerol and measures of reproduction in the first 300 d of lactation. In spite of the associations with disease and survival, increasing concentrations of hepatic triacylglycerol was marginally associated with reproduction. The proportion of cows that received at least one AI tended to be less and pregnancy at first AI tended to decrease as triacylglycerol increased. Nonetheless, the rates of insemination and of pregnancy were not associated with hepatic triacylglycerol. Indeed, when cows were categorized as having concentrations of hepatic triacylglycerol <2.5%, 2.5 to 4.99%, or $\geq$ 5.0%, the hazard of pregnancy was undistinguishable among the

3 groups. It was anticipated that increments in hepatic triacylglycerol, which are often related to smaller NE_L balance, as shown herein, would be associated with reduced pregnancy per AI and rate of pregnancy; however, such associations were not observed in the present study.

A recent meta-analysis by Civiero et al. (2021) showed limited association between energy balance in early lactation, between 4 and 21 DIM, and measures of reproduction in dairy cows in the first 200 DIM. The authors did show an association between energy balance postpartum and estrus with cows in better energy balance having shorter interval to first observed estrus and a larger proportion of cows detected in estrus before 42 DIM. Nevertheless, they were unable to detect associations between energy balance and pregnancy per AI or rate of pregnancy in the first 200 DIM. It is important to note that cows in Civiero et al. (2021) had moderate milk production, with ECM in the first 150 DIM below 30 kg/d. On the contrary, in herds with greater milk production potential, above 35 to 40 kg of ECM/d, then negative NE_L and associated changes in BW and BCS have been linked with increased risk of anovulation (Butler, 2000; Santos et al., 2010, 2016), which delays first insemination and reduces pregnancy per AI (Santos et al., 2016). Because cows received timed AI for first postpartum insemination, such manipulation likely abolished any potential association between hepatic lipidosis and day postpartum at first AI. Indeed, we found no association between hepatic triacylglycerol and DIM at first insemination (data not shown), likely related to the reproductive program imposed with synchronization for first AI. It is also possible that subjecting cows to the double Ovsynch protocol, that was implemented in 3 of the 4 experiments used for this study, minimized any influence of hepatic triacylglycerol on pregnancy per AI at first insemination, as shown for the pregnancy diagnosis on d 74 after AI. In cows subjected to insemination following detected estrus, those with hepatic lipidosis had reduced daily rate of pregnancy (Jorritsma et al., 2000), although the effect was all mediated by reduced insemination rate, and not pregnancy per AI. In Jorritsma et al. (2000), authors classified cows as having hepatic triacylglycerol below or above 5% on a wet basis and observed that cows above the threshold had reduced detection or estrus and hazard of pregnancy, but pregnancy per AI at first insemination was almost the same to that of cows below the threshold. Thus, it is possible that hepatic triacylglycerol is not associated with pregnancy to an insemination, as observed herein and by Jorritsma et al. (2000), and by using timed AI, we abolished many of the potential effects of hepatic lipidosis on rates of

insemination that influence the rate of pregnancy in dairy cows.

Study Limitations

The study incorporated data from 4 experiments in 2 different sites over the course of 9 years. Although case definitions of diseases were the same, the diagnoses were performed by different people and differences might have occurred relative to the consistency of detecting cows with clinical disease. Nevertheless, the range of hepatic triacylglycerol among the different experiments (experiment 1, 0.75 to 9.44%; experiment 2, 0.76 to 13.4%; experiment 3, 0.60 to 12.52%; and experiment 4, 0.44 to 16.06%) included cows with mild, moderate, or exacerbated hepatic lipidosis and potential differences in disease diagnosis is expected to be applied to all ranges of hepatic triacylglycerol. Also, because the random effect of experiment nested within treatment was included in all statistical analyses, it is anticipated that variation among experiments would be accounted by that random effect.

A second potential limitation is the fact that this study used data from 13 experiment-treatment combinations in only 2 farms. Thus, the external validity of the inferences from the data might have restrictions, as any experiment or study that is not multisite. In order to address that limitation, we analyzed the data using the random effect of the interaction between hepatic triacylglycerol and treatment nested within experiment as part of the error term to test the associations of hepatic triacylglycerol and responses analyzed. By applying this error term with the respective denominator degrees of freedom, the detection of statistical significance implies that the associations observed apply to all combinations of treatment-experiments used in the study.

A third limitation of the study is that a single measurement of hepatic triacylglycerol on d 8 postpartum was used. Our assumption is that cows with mild hepatic lipidosis on d 8 also had mild lipidosis before or after d 8, whereas those with exacerbated hepatic lipidosis on d 8 likely had increased hepatic triacylglycerol before or after d 8. This assumption is based on the fact that bovine liver disposes of re-esterified fatty acids slowly because of the limited lipase activity (Hocquette et al., 1998) and rate of lipoprotein secretion (Emery et al., 1992), thus relying mostly on oxidation. Experiments in the literature in which liver tissue was sampled multiple times showed that hepatic triacylglycerol remained elevated and relatively stable in the first 3 to 4 wk of lactation (Hartwell et al., 2000; Zenobi et al., 2018a; Bollatti et al., 2020b).

CONCLUSIONS

Increased infiltration of triacylglycerol into the hepatic tissue is common in early postpartum dairy cows, but the exact threshold of triacylglycerol that compromises performance remains elusive. The current study attempted to understand the relationships between hepatic triacylglycerol in the first 2 wk postpartum and productive, health, and reproductive performance. Morbidity and risk of multiple diseases increased as hepatic triacylglycerol increased, thus not allowing a clear determination of the critical threshold as the associations were linear. Most production data showed quadratic associations with hepatic triacylglycerol and yields of milk and milk components initially increased and then decreased with increments in triacylglycerol, thus suggesting that the degree of hepatic lipidosis to negatively affect production performance to be greater than 4 to 7% on a wet basis. The increments in productive performance observed with increased hepatic triacylglycerol were partially explained by increased tissue catabolism, which might underlie the increased risk of morbidity and multiple diseases observed in this study, although reproduction was not associated with hepatic triacylglycerol. Collectively, hepatic lipidosis beyond 4 to 7% hepatic triacylglycerol was associated with impaired performance, health, and survival.

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